Tianyi Wang

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Research Interests

Post-training algorithms, especially reinforcement learning, and optimizer design and implementation, including PyTorch/TensorFlow reproduction.

Education

- **Beijing University of Posts and Telecommunications** **09/2023 – Expected 07/2027**
  B.S. in Computer Science and Technology *Beijing, China*
  – GPA: 3.76/4.00; major rank: 20/360.
  – CET-6 (503), CET-4 (571).

Research Experience

- **SPPO: Sequence-Level PPO for Long-Horizon Reasoning Tasks** [] [] **04/2026 – ACL 2026 Oral**
  – Recast token-level PPO as a sequence-level contextual bandit and used a decoupled value function for more stable long-horizon reasoning optimization with lower memory and computation cost.
- **Anchored Policy Optimization: Mitigating Exploration Collapse via Support-Constrained Rectification** [] [] **05/2026 – ICML 2026**
  – Proposed APO, replacing global KL matching with support coverage to mitigate exploration collapse and improve Pass@1 and diversity in RLVR.
- **DyJR: Preserving Diversity in Reinforcement Learning with Verifiable Rewards** [] **04/2026 – Under Review**
  – Used a dynamic buffer and Jensen–Shannon regularization to reduce diversity collapse in mathematical reasoning and Text-to-SQL tasks.
- **CAFTBench: A Benchmark for Continuous Fine-Tuning of Tool-Calling Agents** **05/2026 – Under Review**
  – Built a benchmark for continuous fine-tuning of tool-calling agents, covering routing, composition, API upgrades, tool avoidance, and structured-output robustness.
- **Worst-Case Dropout: Learning Task-Critical Binary Masks for Robust Neural Regularization** **05/2026 – Under Review**
  – Formulated dropout as constrained min–max binary-mask learning to improve robust neural regularization.
- **PyTorch Implementation of Spectral-Sphere-Optimizer (SSO)** [] [] **02/2026 – Merged PR**
  – Contributed and merged a PyTorch implementation of SSO for direct training and reproducibility.

School Experience

- **Undergraduate Student, Beijing University of Posts and Telecommunications** **09/2023 – Present**
- **NLP Laboratory, Southern University of Science and Technology** **06/2025 – 06/2026**
- **GAMMA LAB, Beijing University of Posts and Telecommunications** **08/2024 – 04/2025**

Work Experience

- **Tencent YouTu Laboratory** [Mentor: **Zihan Xu** and **Xiaoyu Tan**] **06/2026 – Present**
  Research Intern *Shanghai, China*

Skills

Programming: Python, C, Java **Database:** MySQL